

Math708 – Foundations of Computational Mathematics I

Fall 2011

Homework 2

1. Let a partition $\Delta : a = x_0 < x_1 < \cdots < x_n = b$ of the interval $[a, b]$ as well as support ordinates $f_0, f_1, \dots, f_n \in \mathbb{R}$ be given.
 - a) Show that for each number $f'_0 \in \mathbb{R}$ there exists a unique interpolating quadratic spline s_2 that satisfies the additional condition $s'_2(x_0) = f'_0$. Provide an algorithm to calculate s_2 .
 - b) Find the interpolating quadratic spline s_2 with periodic boundary conditions $s'_2(x_0) = s'_2(x_n)$. Make a statement on existence and uniqueness of s_2 .
2. Let a partition $\Delta : a = x_0 < x_1 < \cdots < x_n = b$ of the interval $[a, b]$ as well as a function $f \in \mathbb{C}^1[a, b]$ be given. A function $s_H \in \mathbb{C}^1[a, b]$ is called the **Hermite** cubic spline interpolating f at $\{x_i\}_{i=0}^n$ if

$$\begin{cases} s_H|_{[x_i, x_{i+1}]} \in \mathbb{P}_3, & i = 0, 1, \dots, n-1 \\ s_H(x_i) = f(x_i), s'_H(x_i) = f'(x_i), & i = 0, 1, \dots, n. \end{cases}$$

Now suppose that f is a polynomial of degree 3. Show that the **Hermite** cubic spline s_H which interpolates f at $\{x_i\}_{i=0}^n$ is identical to f (i.e., $s_H \equiv f$).

3. (**Programming Exercise for Spline**) Write a program to interpolate the function

$$f(x) = \frac{1}{25x^2 + 1}, \quad x \in [-1, 1],$$

using a cubic spline $s_3(x)$ at the equidistant points $x_j = -1 + 2j/n$, $j = 0, 1, \dots, n$. with

- Natural boundary conditions: $s''_3(-1) = s''_3(1) = 0$.
- Periodic boundary conditions: $s'_3(-1) = s'_3(1)$ and $s''_3(-1) = s''_3(1)$.

Plot the graphs and calculate $\|s_3 - f\|_\infty$ respectively for $n = 5, 10$. Also submit your codes in both paper and email.

4. For even N , let $(N+1)$ support points $a \leq x_0 < x_1 < \cdots < x_N < a + 2\pi$ and support ordinates $f_0, \dots, f_N \in \mathbb{C}$ be given. Show that:
 - a) There is exactly one trigonometric polynomial of the form

$$T(x) = A_0/2 + \sum_{j=1}^{N/2} (A_j \cos(jx) + B_j \sin(jx))$$

with $T(x_k) = f_k$ for $k = 0, 1, \dots, N$.

- b) If all the support ordinates $f_0, \dots, f_N$ are real, the coefficients A_j, B_j are also real.

5. Let $N \in \mathbb{N}$ be even number. For $\{f_k\}_{k=0}^{N-1} \in \mathbb{R}^N$, suppose that $f_{N-k} = f_k$ for $k = 1, \dots, N/2 - 1$. Show that:

$$d_j \in \mathbb{R} \quad \text{for } j = 0, \dots, N-1, \quad \text{and} \quad d_{N-j} = d_j \quad \text{for } j = 1, \dots, N-1.$$

6. (**Programming Exercise for FFT**) For any function $f(x)$ defined on $[0, L]$, there exists an interpolation of f using the real trigonometric polynomials of the form

$$T(x) = A_0 + 2 \sum_{k=1}^{N/2-1} \left((A_k \cos(\frac{k2\pi x}{L}) + B_k \sin(\frac{k2\pi x}{L})) \right) + A_{N/2} \cos(\frac{N\pi x}{L})$$

with even number N and equidistant support points $x_j = jL/N$ for $j = 0, 1, \dots, N-1$. The coefficients A_k, B_k can be determined indirectly through the fast Fourier transform (See the lecture notes). Interpolate the following two functions

- $f(x) = x^2(1-x)^2, \quad x \in [0, 1]$.
- $f(x) = e^{2x}, \quad x \in [0, 2]$.

with $N = 2^m$, $m = 2, 4, 6$. Plot the figures respectively and calculate the maximum of absolute occurring errors at the points (to be linearly transformed) $x_j = jL/40$ for $j = 1, 2, \dots, 40$.